

Microprocessors :

- **The Microprocessor.** Rather than implement a rigid set of logic equations, the microprocessor fetches, from a memory unit, “instructions” about the problem to be solved. These instructions are called the control program.

Microcomputer System Basics :

- Microcomputer is Built around a **microprocessor**.
- It Includes:
 - **Memory** (stores program + data)
 - **Input devices** (like switches, sensors)
 - **Output devices** (like LEDs, displays, motors).

Example Given :

- Input → a switch (e.g., smoke detector or high-temperature sensor).
Microprocessor runs a **control program** stored in memory.
Output → an LED warning light turns ON when condition is met.
- **Why Software Control is Powerful**
- In old digital designs (without microprocessors), if you wanted to change functionality → you had to **redesign the hardware circuit** (expensive & time-consuming).
- With a microprocessor → functionality comes from the **program** (software).
- To add/change features → just update the program, hardware stays the same

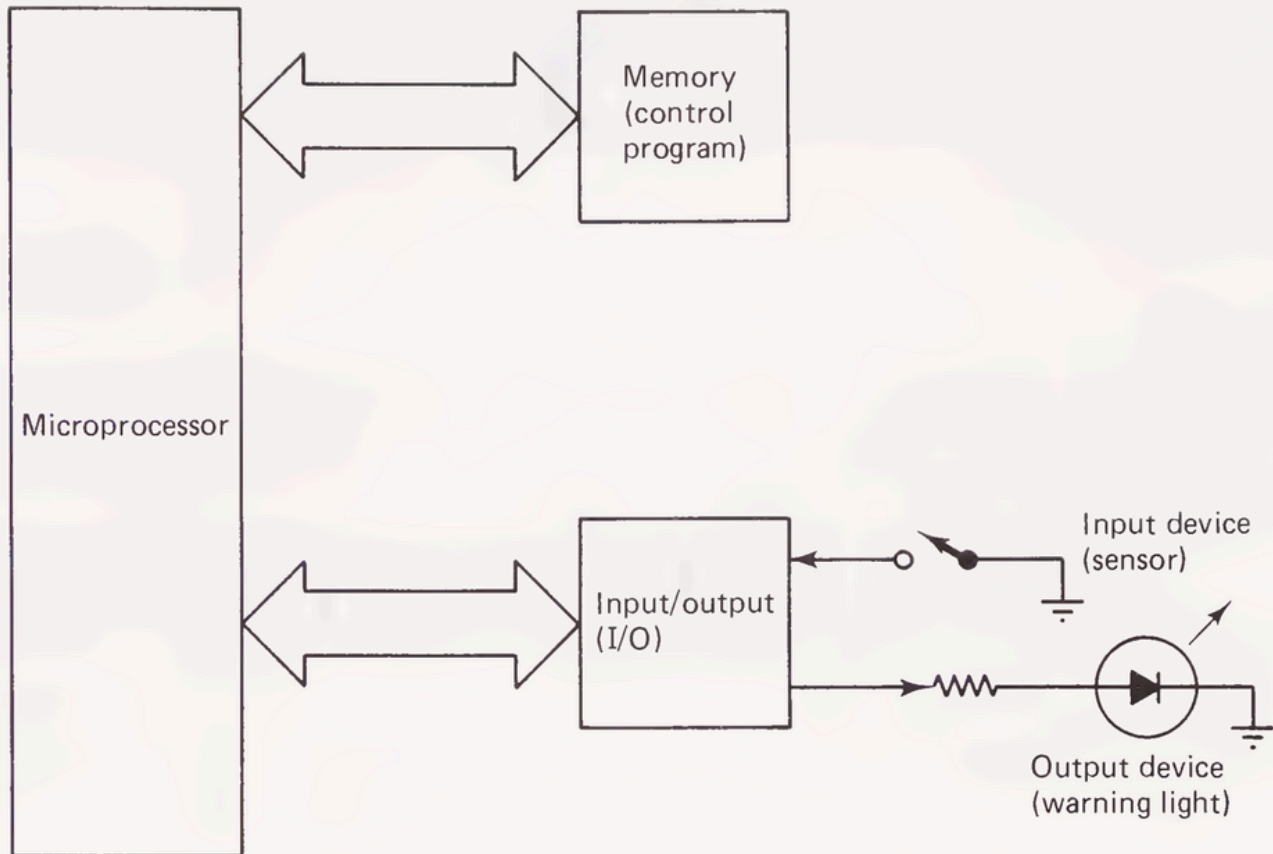


Figure 1.1 The microprocessor has revolutionized the digital design process. Combined with a memory unit and a set of I/O devices, a microcomputer is formed.

- The 8086 microprocessor is actually just one integrated-circuit (IC) chip in a family of 16-bit microprocessors designed by Intel.

Stored Program Computers :

- **John von Neumann** proposed a totally new computer architecture. Now called the **stored program** concept, von Neumann suggested that just like the data, the **computer program be stored in memory. The computer would then be permanently wired to fetch its instructions from memory instead of being rewired for each new program**

- Interestingly enough, virtually all computers today are still based on the **von Neumann architecture** and stored program concept.